

Highlights

HG-Mem: Hierarchical Graph and Multi-Path Retrieval based Long-Term Memory Management for Conversational Agents

Guo Yuhang, Fan Yanfang, Chen Ruoyu, Cai Ying, Wang Jingqi, Li Haitao, He Xiaoye, Ji Zhenghao

- HG-Mem builds a hierarchical memory graph for long-term memory management.
- HG-Mem proposes multi-path retrieval with multi-signal ranking for fine-grained evidence localization.
- The training-free retrieval method introduces no additional LLM calls, while maintaining low retrieval latency.
- HG-Mem improves retrieval and end-to-end question answering over existing methods on LoCoMo and LongMemEval datasets.

HG-Mem: Hierarchical Graph and Multi-Path Retrieval based Long-Term Memory Management for Conversational Agents

Guo Yuhang^a, Fan Yanfang^{a,*}, Chen Ruoyu^a, Cai Ying^a, Wang Jingqi^a, Li Haitao^a, He Xiaoye^a, Ji Zhenghao^a

^a*College of Computer Science, Beijing Information Science & Technology University, Beijing, 102206, China*

Abstract

As large language model (LLM)-driven conversational agents are increasingly deployed in sustained long-term interactions, accurately identifying original evidence relevant to the current query from continuously expanding dialogue history has become a core challenge for conversational agents. Existing methods typically retrieve memory representations of different granularities as independent candidate units, they do not explicitly model hierarchical entailment relationships, which limits their ability to leverage high-level semantic information for fine-grained evidence localization. Lacking the ability to pinpoint fine-grained evidence, the model is forced to rely on coarse-grained sessions to generate answers, rendering it unable to precisely capture the specific factual fragments actually required to answer the question. To address this limitation, we propose Hierarchical Graph Memory (HG-Mem), a hierarchical graph and multi-path retrieval based framework tailored for long-term memory management. HG-Mem organizes dialogue history into a hierarchical memory graph consisting of session, topic-cluster, and turn nodes. Specifically, topic-cluster nodes are constructed by applying hierarchical agglomerative clustering to turns within each session. Two fundamental entailment relationships, namely session turn and topic-cluster turn, bridge high-level semantic nodes and their underlying original turn evidence. During the retrieval stage, HG-Mem collects candidate turns through

*Corresponding author.

Email address: fyfhappy@bistu.edu.cn (Fan Yanfang)

three complementary paths, namely Best Matching 25 (BM25) path, the session path, and the topic-cluster path. The merged candidate set is further ranked via multi-signal ranking, which integrates BM25 lexical scores, turn-level semantic scores, and topic-aware semantic scores. Extensive experiments on the LoCoMo and LongMemEval benchmarks verify its superior performance in both memory retrieval and end-to-end question answering, especially achieving prominent improvements in relevant evidence coverage under large retrieval depths. Compared with the optimal baseline methods on each dataset, HG-Mem improves Recall@10 by 7.00 and 1.99 percentage points and end-to-end F1-score gains of 3.07 and 1.33 percentage points, respectively. Ablation experiments and weight sensitivity analyses further validate the complementarity of the multi-path recall strategies and multiple ranking signals. Summary generation, topic extraction, and clustering are all performed in the offline construction stage, so HG-Mem maintains low online retrieval latency. **Code:** <https://github.com/Rsir-i/HG-Mem>

Keywords:

conversational agent, long-term conversational memory, hierarchical graph, multi-path retrieval

1. Introduction

LLM-driven conversational agents are evolving from single-turn question answering toward sustained long-term interactions. Beyond generating high-quality responses, such systems must continuously accumulate, organize, and exploit information over time. However, since LLMs lack persistent state storage, long-term interactions suffer from lost conversational history and user preferences, as well as impaired contextual consistency. Unlike short multi-turn conversation, real interactions between users and systems often span days or longer, during which new tasks, user behavioral preferences, and contextual information continually emerge. Consequently, an effective mechanism for preserving, organizing, and accessing historical information is essential to improving agent autonomy, personalization, and reliability over long-term interactions[1, 2, 3].

As the history grows, simply retaining more information does not guarantee that the model can use it effectively. Retrievable external memory provides a scalable alternative by organizing dialogue history into retrievable memory units and dynamically retrieving relevant content when needed[7, 8,

9, 10]. Since a current query usually depends on only a subset of the history, the core challenge for long-term conversational memory management has shifted from storing more history to accurately locating evidence relevant to the current task within complex interactions. Feeding the entire history directly to an LLM is constrained by context-window size and inference cost[4], and also introduces substantial irrelevant noise[5, 6].

Long-term dialogue history, however, typically contains multiple sessions, each comprising continuous interactions centered on different topics. Therefore, semantic information at varying levels of granularity is presented across the session, topic, and turn levels. Retrieving these representations as independent memory units severs the hierarchical relationships among them, preventing session-level and topic-level semantics from effectively guiding the localization of relevant evidences, and thereby limiting accurate access to the original history[11].

To improve the organization and utilization of long-term dialogue history, prior studies have explored memory representations at turn-level, topic-level, and session-level granularities. Each granularity offers different trade-offs in information completeness, semantic abstraction, and retrieval scope. Nevertheless, long-term conversational memory retrieval must still reconcile the preservation of original factual evidence with the effective use of high-level semantic information[12].

Research on long-term conversational memory has recently progressed beyond simple history storage and summary compression toward structured memory organization and multi-granularity retrieval. External memory and recursive summarization methods have shown that storing dialogue history outside the model context and retrieving or compressing it on demand can alleviate context-window limitations. Meanwhile, benchmarks such as LoCoMo[13] and LongMemEval[14] have enabled more systematic assessment of cross-session question answering, temporal reasoning, knowledge updating, and the use of historical information. Accordingly, recent work has investigated topic-segmented memory construction, dynamically-linked agent memory, and mechanisms for associating and selecting memories at multiple granularities[15, 16, 17]. Collectively, these studies suggest that long-term conversational memory should be organized as a structured, multi-granularity system rather than as a collection of independent text fragments.

To address this challenge, we propose HG-Mem, a hierarchical graph and multi-path retrieval based framework tailored for long-term memory management. HG-Mem organizes dialogue history as a hierarchical memory graph

containing session, topic-cluster, and turn nodes, and exploits entailment relationships to connect high-level semantic representations with original turn-level evidence. Turn nodes always constitute the final retrieval output, whereas session and topic-cluster nodes guide candidate localization.

At retrieval time, HG-Mem employs a hierarchical-graph-based, three-path candidate recall mechanism followed by multi-signal ranking over the merged candidate set. This design allows memories at different granularities to participate in retrieval while ensuring that every final result is grounded in an original turn. Our main contributions are as follows:

- (1) We introduce a hierarchical graph representation for long-term conversational memory that organizes dialogue history into session, topic-cluster, and turn nodes and explicitly models entailment relations linking high-level memory nodes to original turn nodes.
- (2) We develop a hierarchical-graph-based, training-free memory retrieval framework that maps multi-granularity retrieval results to original turn-level evidence through three-path candidate recall and multi-signal ranking.
- (3) We evaluate HG-Mem on the LoCoMo and LongMemEval long-term conversational memory benchmarks, demonstrating its strong effectiveness and high runtime efficiency for long-term dialogue memory modeling, as well as the necessity of our core hierarchical graph-based memory design.

2. Related Work

2.1. Long-Term Conversational Memory Evaluation Benchmarks

Unlike conventional short-text dialogue, long-term conversational memory requires a model to preserve, organize, and retrieve task-relevant information across extensive interaction history. A suite of benchmarks has recently been introduced to systematically evaluate these capabilities. Maharana et al.[13] proposed LoCoMo, which assesses information retrieval, event understanding, and temporal reasoning across long time spans and multiple sessions, using context-dependent questions to test how well models exploit historical information. Wu et al.[14] further proposed LongMemEval, which evaluates an agent’s long-term memory from multiple perspectives, including information extraction, cross-session reasoning, temporal reasoning, knowledge updating, and user-preference retention.

These benchmarks demonstrate that the core challenge of long-term conversational memory lies not simply in expanding the context window, but in

accurately locating task-relevant memories within continuously growing dialogue history. Designing effective memory organization strategies and precise retrieval mechanisms has therefore become a central research direction in long-term conversational memory.

2.2. Single-Granularity Memory Management

Early studies on long-term conversational memory mainly focused on storing dialogue history outside the model context and retrieving them on demand. These conventional approaches generally follow a single-granularity memory paradigm, converting historical interactions into uniformly structured memory units.

Zhong et al.[18] proposed MemoryBank, which enables a model to maintain user-related information through long-term storage, updating, and forgetting mechanisms and to revise its memory state as interactions continue. Packer et al.[19] proposed MemGPT, which draws on hierarchical storage in operating systems, treats the limited context window as working memory, and extends the accessible information space through external storage. In addition, Wang et al.[20] proposed the recursive summarization strategy, which compresses historical interactions into high-level semantic representations, reducing the cost of maintaining long contexts.

Although these approaches alleviate the context-window limitations of LLMs, relying solely on a single memory granularity fails to simultaneously preserve semantic completeness and concrete factual evidence. Fine-grained memory retains original dialogue details but lacks high-level organizational structures, whereas summary-level memory achieves higher semantic compression yet may discard specific details, such as temporal information and event specifics. Consequently, single-granularity memory paradigms cannot simultaneously satisfy the demands of holistic background understanding and precise factual localization in long-term conversations.

2.3. Multi-Granularity Memory Management

To address the limitations of single-granularity representations, recent studies have adopted multi-granularity memory organizations that preserve dialogue information at distinct abstraction levels and enhance the overall effectiveness of long-term memory retrieval.

Pan et al.[15] proposed SeCom, pointing out the drawbacks of existing turn-level, session-level, and summarization-based memory paradigms. The

model constructs segment-level memory by partitioning dialogues into topically coherent segments and leverages compression for memory denoising, revealing that memory-unit granularity plays a vital role in long-term memory retrieval

Xu et al.[16] proposed A-Mem, a structured memory organization framework inspired by the Zettelkasten knowledge-management system. It represents historical information as memory notes with attribute descriptions and associative relations, and builds an evolving memory network via dynamic linking mechanisms.

Xu et al.[17] proposed MemGAS, which focuses on multi-granularity memory association and selection. It jointly models memory units at different granularities, employs a Gaussian mixture model to probabilistically characterize cross-granularity associations, and propagates associative signals through Personalized PageRank.

These studies demonstrate that multi-granularity memory enables comprehensive representation of long-term dialogue history from complementary semantic perspectives. However, existing methods generally treat memory units of different granularities as related but largely independent memory units, lacking explicit modeling of hierarchical entailment relationships. Furthermore, some approaches rely on dynamic memory updates, probabilistic association modeling, and additional reasoning procedures, which substantially increase computational overhead and costs. In contrast, HG-Mem organizes multi-granularity memories through a hierarchical-graph-based structure and anchors all retrieved results to verifiable original turn-level evidence.

2.4. Graph-Structured Memory for Retrieval-Augmented Generation

With advances in knowledge graphs and graph-enhanced retrieval, researchers have increasingly used graph structures to model complex information relations. Edge et al.[21] proposed GraphRAG, which builds entity-relation graphs and community structures, extending conventional vector retrieval to graph-based global information retrieval and enhancing the modeling of complex knowledge relations. Gutiérrez et al.[22] proposed HippoRAG, which draws on memory mechanisms inspired by cognitive science and uses knowledge graphs to emulate long-term associative memory, thereby supporting multi-hop relational retrieval.

Other studies have explored graph-structured memory for agents, using entity, event, or relation nodes to establish flexible associations. Existing

graph retrieval methods, however, primarily target knowledge-relation modeling or entity-association propagation rather than the hierarchical organization of multi-granularity long-term conversational memory. Nodes in a knowledge graph typically represent entities or factual relationships, whereas long-term dialogue history is characterized by continuous interactions and contextual dependencies, and historical information spanning different scopes often conveys semantics at varying levels of abstraction. Consequently, structured representations tailored to the inherent nature of dialogue memory are required, highlighting the clear distinctions between the two. Unlike graph retrieval methods that primarily model entity relations or external knowledge associations, HG-Mem focuses on entailment among memories at different granularities within long-term dialogue. Its graph structure organizes historical information and constrains candidate expansion rather than performing knowledge reasoning through a generic graph-propagation mechanism.

3. Method

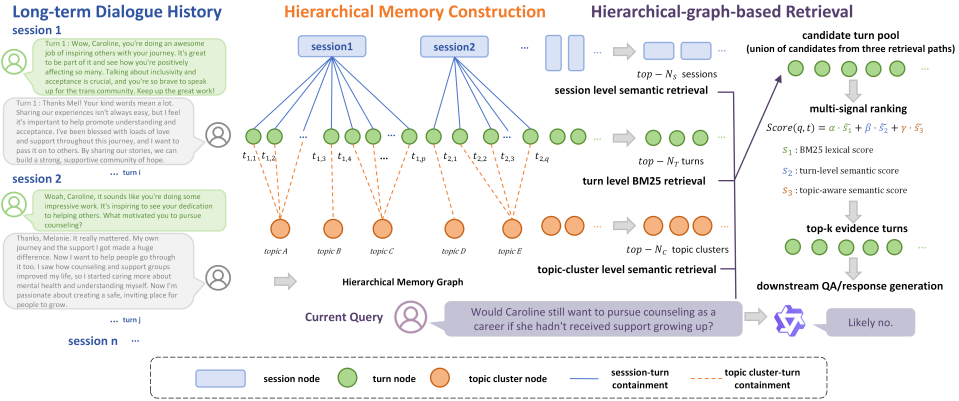


Figure 1: Overall framework of our HG-Mem

As illustrated in Figure 1, HG-Mem is a hierarchical graph and multi-path retrieval based framework tailored for long-term memory management. The framework consists of two major stages:

(1) Hierarchical memory construction: the method builds a hierarchical memory graph from raw dialogue history using session, topic-cluster, and turn nodes.

(2) Hierarchical-graph-based retrieval: given a user query, candidate evidence turns are recalled through three channels—direct BM25 matching between the query and dialogue turns, and semantic embedding-based cosine similarity matching with session paths and topic-cluster paths followed by mapping to corresponding turns via intrinsic associations—then re-ranked using multiple signals and returned as the final top- k results.

3.1. Definition

The goal of long-term conversational memory retrieval is to identify the historical evidence relevant to the current query within a continuously growing interaction history. Given a user’s long-term dialogue history $H = \{S_1, S_2, \dots, S_n\}$, where S_i denotes the i -th historical session. Each session consists of multiple consecutive interaction turns $S_i = \{t_{i,1}, t_{i,2}, \dots, t_{i,j}\}$, where $t_{i,j}$ denotes the j -th turn in the i -th session. Given a current query q , the long-term conversational memory retrieval task is formulated as $\varepsilon_q = R(q, H)$, where $R(\cdot)$ denotes the memory retrieval function, and ε_q is the returned set of top- k evidence turns. Because session summaries and topic representations are compressed or aggregated, they are not directly suitable as verifiable factual evidence. We therefore use original turns as the final retrieval units. Session and topic-cluster nodes are not directly included in the final output. Instead, they facilitate candidate-turn generation by leveraging hierarchical memory relationships.

3.2. Hierarchical Memory Construction

3.2.1. Definition of the Hierarchical Graph

We represent a long-term dialogue history as a hierarchical graph:

$$G = (V, E), \quad (1)$$

where V denotes the node set, and E denotes the set of edges among nodes. The node set contains three types of memory nodes:

$$V = V_S \cup V_C \cup V_T, \quad (2)$$

where V_S is the set of session nodes, V_C is the set of topic cluster nodes, and V_T is the set of turn nodes.

3.2.2. Construction of Session and Turn Nodes

For each session S_i , we use an LLM to generate its summary d_i . A semantic encoder $\phi(\cdot)$ then produces the session-level semantic vector $e_i^S = \phi(d_i)$. The session node is therefore defined as $v_i^S = (d_i, e_i^S)$, where the session node represents the global semantic context of the complete session.

For each original turn $t_{i,j}$, HG-Mem retains its original text as final evidence and computes its semantic representation $e_{i,j}^T = \phi(t_{i,j})$. The turn node is therefore defined as $v_{i,j}^T = (t_{i,j}, e_{i,j}^T)$.

3.2.3. Construction of Topic-Cluster Nodes

To establish local semantic organization within each session, we constructed topic-cluster nodes in HG-Mem. For each turn $t_{i,j}$, an LLM first generates a corresponding topic description $z_{i,j}$. To prevent topic confusion across sessions, topic clustering is performed independently within each session. The topics are then grouped using hierarchical agglomerative clustering with cosine distance[23]. Let $\pi_{i,j}$ denote the topic-cluster index of turn $t_{i,j}$. The r -th topic-cluster is represented as:

$$C_{i,r} = \{z_{i,j} \mid \pi_{i,j} = r\}. \quad (3)$$

Each topic cluster corresponds to a topic-cluster node, whose semantic representation is obtained by averaging the topic embeddings of all turns in the cluster:

$$e_{i,r}^C = \frac{1}{|C_{i,r}|} \sum_{t_{i,j} \in C_{i,r}} \phi(z_{i,j}), \quad (4)$$

The topic-cluster node is therefore defined as $v_{i,r}^C = (e_{i,r}^C)$. A topic-cluster node captures local topical continuity within a session.

3.2.4. Construction of Edges

HG-Mem contains two types of edges. First, a session-turn edge connects a session node to every turn node it contains:

$$E_{S-T} = \left\{ (v_i^S, v_{i,j}^T) \mid t_{i,j} \in S_i \right\}. \quad (5)$$

Second, a topic-cluster turn edge connects a topic-cluster node to each turn node it associates:

$$E_{C-T} = \left\{ (v_{i,r}^C, v_{i,j}^T) \mid t_{i,j} \in C_{i,r} \right\}. \quad (6)$$

The final edge set is:

$$E = E_{S-T} \cup E_{C-T}. \quad (7)$$

Through these two relations, HG-Mem organizes memory at different granularities into a unified hierarchical memory graph.

3.3. Hierarchical-graph-based Retrieval

3.3.1. Query-Driven Three-Path Candidate Recall

HG-Mem performs three-path candidate recall over the hierarchical memory graph. Session and topic-cluster nodes are not final outputs; instead, their hierarchical connections map high-level semantic abstraction to specific dialogue contents.

First, the BM25 path performs lexical retrieval over the content of each turn node, the result is represented as:

$$s_{\text{bm25}}(q, v_{i,j}^T) = \text{BM25}(q, t_{i,j}), \quad (8)$$

we select the top- N_T turn nodes with the highest s_{bm25} , denoted by C_T . This path recalls original turns that directly match the query lexically.

Second, the session path computes the semantic similarity between each session node v_i^S and the query:

$$s_{\text{session}}(q, v_i^S) = \cos(\phi(q), e_i^S), \quad (9)$$

we select the top- N_S session nodes with the highest s_{session} and expand their contained turns through E_{S-T} , yielding C_S . This path uses session-level global semantics to identify the relevant span of the dialogue history.

Finally, the topic-cluster path computes the semantic similarity between each topic-cluster node $v_{i,r}^C$ and the query:

$$s_{\text{topic}}(q, v_{i,r}^C) = \cos(\phi(q), e_{i,r}^C), \quad (10)$$

we select the top- N_C topic-cluster nodes with the highest s_{topic} and obtain their corresponding turns through E_{C-T} , yielding C_C . Unlike expansion of an entire session, the topic path expands only a local topical region, reducing the introduction of irrelevant history.

The results from the three paths are merged and deduplicated to form the unified candidate set $C_q = C_T \cup C_S \cup C_C$.

3.3.2. Multi-Signal Evidence Ranking

For the candidate set C_q , HG-Mem jointly ranks candidate turns from complementary relevance perspectives. We collectively refer to the BM25 lexical score, turn-level semantic score, and topic-aware semantic score computed for each candidate as ranking signals. For a candidate turn $t_{i,j} \in C_q$, HG-Mem computes three signals. The first is the BM25 lexical signal:

$$s_1(q, v_{i,j}^T) = \text{BM25}(q, t_{i,j}), \quad (11)$$

which measures the lexical match between the query and the candidate turn.

The second is the turn-level semantic signal:

$$s_2(q, v_{i,j}^T) = \cos(\phi(q), e_{i,j}^T), \quad (12)$$

which measures semantic relevance between the query and the original text of the candidate turn.

The third is the topic-aware semantic signal:

$$s_3(q, v_{i,j}^T) = \cos(\phi(q), e_{i,r}^C), \quad (13)$$

which measures the semantic relevance between the query and the topic-cluster to which the candidate turn belongs.

If a turn lacks a valid topic-cluster representation, its topic-aware semantic signal is set to 0. Because the three signals originate from different sources and have different numerical scales, each signal is normalized within the candidate set. For any ranking signal $s_m(q, t_{i,j})$, where $m \in \{1, 2, 3\}$, the normalized value is:

$$\tilde{s}_m(q, t_{i,j}) = \frac{s_m(q, t_{i,j}) - \min_{t'_{i,j} \in C_q} s_m(q, t'_{i,j})}{\max_{t'_{i,j} \in C_q} s_m(q, t'_{i,j}) - \min_{t'_{i,j} \in C_q} s_m(q, t'_{i,j}) + \epsilon}, \quad (14)$$

where ϵ is a small constant that prevents a zero denominator. The final score assigned to each candidate turn is defined as:

$$\text{Score}(q, t_{i,j}) = \alpha \tilde{s}_1(q, t_{i,j}) + \beta \tilde{s}_2(q, t_{i,j}) + \gamma \tilde{s}_3(q, t_{i,j}), \quad (15)$$

where α , β , and γ are the weights of the three ranking signals and satisfy $\alpha + \beta + \gamma = 1$. HG-Mem ranks the turns by their combined scores and returns the top- k turns as long-term memory evidence.

3.4. Time-Complexity Analysis

HG-Mem concentrates its computational overhead entirely on offline memory construction and thus eliminates the need for parameter optimization during online retrieval. Specifically, the offline preprocessing runtime is predominantly governed by two resource-intensive procedures: the generation of condensed session-level and topic-cluster summaries, and the hierarchical aggregation of dialogue turns into structured topic clusters. These two modules collectively constitute the major preprocessing burden, whereas inverted index construction introduces only marginal computational overhead. In contrast, the online inference pipeline maintains high efficiency by conducting lightweight lexical matching at the turn level and semantic retrieval at both the session and topic-cluster granularities. After aggregating candidate results from the three complementary retrieval branches, HG-Mem calculates unified ranking signals and rearranges the candidate list for final selection. By migrating the most time-consuming summary generation and topic clustering steps to one-time offline preprocessing, our method can effectively alleviate per-query computational pressure and promises consistently low inference latency in practical deployment.

4. Experiments

We conduct experiments on two public long-term conversational memory datasets to evaluate HG-Mem. Our empirical study aims to answer four research questions: (1) Can HG-Mem improve the recall of relevant evidence from long dialogue history? (2) Do three-path candidate recall and multi-signal ranking yield complementary performance benefits? (3) Does HG-Mem achieve consistent performance across different question types? (4) Can HG-Mem maintain low online query overhead after introducing the hierarchical memory structure?

We evaluate overall retrieval and generation performance, weight sensitivity, runtime efficiency, ablation variants, and model performance across various question types.

4.1. Datasets

We use LoCoMo and LongMemEval as our primary experimental datasets. LoCoMo is designed for evaluating conversational memory across long time spans and multi-session dialogue history. Its questions cover personal experiences, user preferences, event information, temporal relations, and cross-turn

reasoning. We take the historical turns annotated in the evidence field as gold-standard evidence, and evaluate whether the retrieved results can cover these relevant original turns. When a question contains multiple gold turns, all annotated relevant turns collectively constitute its evidence set.

For extended dialogue history, LongMemEval targets information extraction, temporal reasoning, cross-session reasoning, knowledge updating, and user-preference memory. We adopt the `longmemeval_s_cleaned` version, and map messages marked with `has_answer=true` to their corresponding turn-level evidence. LongMemEval comprises single-user, single-assist, knowledge, temporal, multi-session, and single-pref memory scenarios, which allows us to analyze the applicability of different candidate-recall paths across diverse question categories.

4.2. Baselines

We compare HG-Mem against the following baseline methods:

BM25[24]: Performs direct lexical matching over all raw turn nodes. It achieves strong exact-match performance for entities, times, locations, numbers, and proper nouns, and serves as a key baseline for long-term conversational memory retrieval.

Similarity[25]: A standard dense-retrieval strategy. A semantic encoder transforms the input query and each historical turn into vector representations, and turns are ranked according to cosine similarity.

RRF[26]: Runs BM25 and turn-level semantic retrieval independently, and fuses their ranked results via Reciprocal Rank Fusion (RRF). This baseline evaluates a straightforward hybrid of lexical and semantic retrieval.

MemGAS[17]: A multi-granularity memory association and selection approach. It selects long-term memory by leveraging multi-granularity memory units, probabilistic distribution modeling, and Personalized PageRank.

A-MEM[16]: A dynamic agent-memory method. It organizes long-term dialogue history by generating memory attributes, constructing memory links, and evolving the memory structure.

SeCom[15]: A memory construction and retrieval approach for personalized conversational agents. It builds segment-level memory via topic segmentation and mitigates historical noise through compression.

For baselines that do not natively output turn-level evidence, we map each retrieved segment, summary, or memory unit back to its corresponding original turns. All approaches are evaluated under identical turn-level

gold-standard evidence and top- k settings. HG-Mem also outputs raw dialogue turns, which guarantees consistent evidence granularity across all compared methods.

4.3. Evaluation Metrics and Experimental Settings

4.3.1. Evaluation Metrics

We evaluate all methods from two perspectives: evidence retrieval and end-to-end question answering. Retrieval performance is measured using Recall@1, Recall@3, Recall@5, Recall@10, and Mean Reciprocal Rank (MRR). Recall@ k quantifies the coverage of relevant evidence within the top- k retrieved results. For a query with multiple gold-standard turns, it corresponds to the fraction of gold turns that are successfully retrieved. MRR measures the reciprocal rank of the highest-ranked relevant turn and is defined as:

$$MRR = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{rank_q} \quad (16)$$

where $rank_q$ denotes the rank of the highest-ranked relevant turn retrieved for question q . If no relevant evidence is present in the returned list, the reciprocal rank for that query is set to 0.

For end-to-end question answering, we adopt the F1-score metric. The top-10 retrieved turns are concatenated to form the external memory context and fed into the generation model. The F1-score is computed based on token-level overlap between generated answers and reference answers. This metric reflects how retrieved evidence influences the quality of downstream generated answers.

4.3.2. Experimental Settings

HG-Mem is implemented in PyTorch. All experiments are conducted on a server equipped with a 14-core AMD EPYC 7453 CPU and an NVIDIA RTX 4090 GPU with 24 GB of VRAM. We employ all-MiniLM-L6-v2 as the semantic encoder to produce 384-dimensional embedding vectors. The maximum input sequence length is set to 512 tokens with a batch size of 32 texts. We apply attention-mask-weighted mean pooling to the final hidden states. Cosine similarity is adopted to compute semantic relevance between the query and turns, topic clusters, as well as sessions.

For LoCoMo, we directly adopt the session summaries provided by the original dataset. For LongMemEval, session summaries are generated offline

via a locally deployed Qwen2.5-7B-Instruct model. Each summary consists of 3–5 sentences, and the prompt is designed to retain key entities from the source dialogue. Turn-level topics for both datasets are also produced by Qwen2.5-7B-Instruct. Each topic is limited to 5–10 words, and the prompt encourages the model to capture core information from the corresponding turn. Turns belonging to the same session are processed in batches of up to 20. Generated session summaries and turn topics are cached locally and reused across subsequent experimental runs.

Topic-clusters are built within each individual session only. MiniLM first encodes the turn topics into embeddings. We then perform average-linkage hierarchical agglomerative clustering based on cosine distance. We do not predefine the number of clusters, fix the distance threshold to 0.55, and permit the existence of isolated topic-clusters. Each topic-cluster is represented by the mean vector of all its turn-topic embeddings, and the longest topic phrase inside the cluster is taken as its textual description.

Regarding three-path candidate recall: the BM25 path retrieves the top-100 turns; the session path selects the top-3 most semantically similar sessions and expands all their constituent turns; the topic-cluster path picks the top-5 most semantically similar topic-clusters and expands their contained turns. After merging and deduplicating candidates from the three paths, each ranking signal is min-max normalized within the candidate pool, and the final ranking score is computed via linear weighted fusion. We grid-search the three weights over $[0, 1]$ with a step size of 0.1, under the constraint that their sum equals 1, which yields 66 valid weight combinations. Recall@10 is used to pick the optimal weight configuration. The final retrieval output length is set to 10 for LoCoMo and 20 for LongMemEval. For end-to-end question answering, the top-10 retrieved turns are fed as context for both datasets. Qwen2.5-7B-Instruct acts as the answer-generation model with a maximum generation length of 100 tokens, and it is prompted to output concise answers only.

4.4. Overall Results

Tables 1 and 2 report retrieval and end-to-end question-answering results on LoCoMo and LongMemEval, respectively. On LoCoMo, HG-Mem obtains the best performance across all metrics. The most prominent improvements are observed in Recall@5, Recall@10, MRR, and F1-score, while Recall@1 remains comparable to BM25. These observations suggest that HG-Mem

Table 1: Experimental comparison of retrieval and generation metrics for different methods on the LoCoMo dataset (all metrics are reported as percentages).

Method	R@1	R@3	R@5	R@10	MRR	F1-score
BM25 [24]	29.93	44.95	51.89	58.72	42.38	32.71
Similarity [25]	14.13	26.71	33.49	42.90	25.47	28.30
RRF [26]	20.73	36.75	44.53	54.54	34.47	31.40
MemGAS [17]	4.23	12.89	19.59	28.38	12.56	10.23
A-Mem [16]	19.08	32.88	41.04	52.17	28.46	28.54
SeCom [15]	3.55	8.72	14.12	23.43	9.39	9.36
HG-Mem	30.06	49.44	56.88	65.72	45.41	35.78

mainly boosts the coverage of relevant evidence and overall ranking quality, instead of consistently lifting the top-ranked result for every query.

Table 2: Experimental comparison of retrieval and generation metrics for different methods on the longmemeval_s_cleaned dataset (all metrics are reported as percentages).

Method	R@1	R@3	R@5	R@10	MRR	F1-score
BM25 [24]	35.26	60.69	69.83	81.20	65.93	42.22
Similarity [25]	26.29	46.83	57.63	74.27	54.98	40.83
RRF [26]	30.42	56.16	68.80	83.71	62.27	42.92
MemGAS [17]	24.82	43.54	49.93	74.60	57.65	16.33
A-Mem [16]	29.00	54.20	64.60	70.40	43.40	6.02
SeCom [15]	21.59	31.47	37.70	43.51	47.88	9.95
HG-Mem	35.32	61.80	72.83	85.70	66.80	44.25

A similar trend can be seen on LongMemEval. HG-Mem yields larger performance gains at higher retrieval depths and also improves end-to-end question answering. This demonstrates that more complete coverage of relevant evidence brings benefits to downstream answer generation. Across both datasets, the main improvements of HG-Mem stem from enhanced evidence coverage for long dialogue history and better candidate-ranking quality.

4.5. Weight Sensitivity Analysis

To analyze how the three ranking signals affect final retrieval accuracy, we conduct a grid-search over weights for the BM25 lexical score, turn-level semantic score, and topic-aware semantic score. The three-path candidate-recall

strategy is kept fixed, and we evaluate the performance under different weight combinations on LoCoMo and LongMemEval.

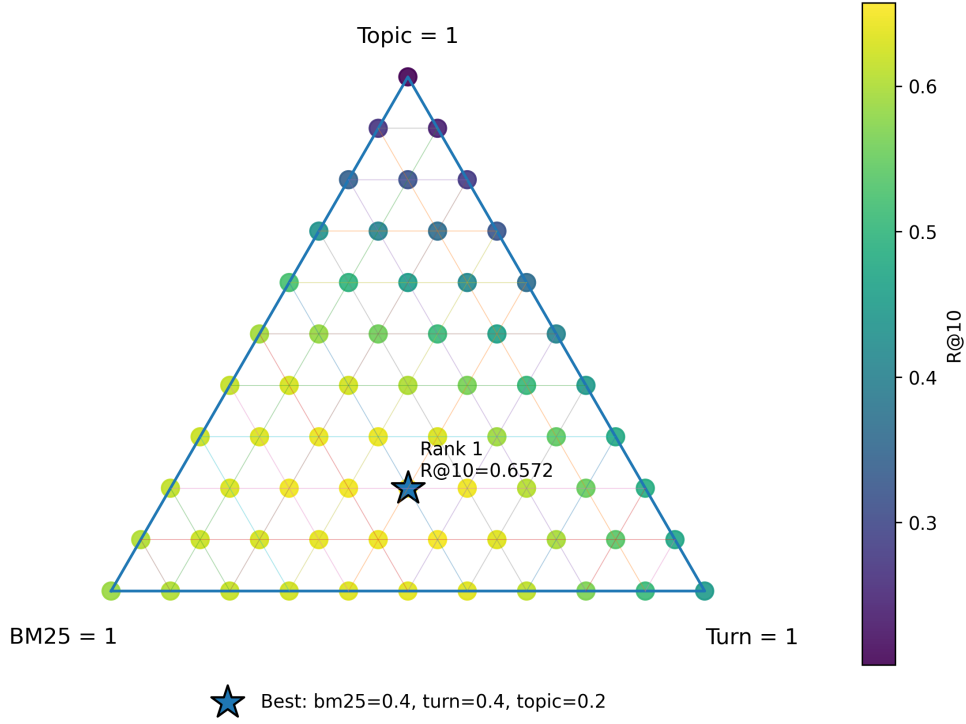


Figure 2: Grid-search results for the three ranking weights on LoCoMo colored by R@10

Figures 2 and 3 report the weight-search results on LoCoMo, while Figures 4 and 5 show the corresponding results on LongMemEval. The optimal weight values for LoCoMo are 0.4, 0.4, and 0.2. This demonstrates that the BM25 lexical score and turn-level semantic score contribute comparably to final ranking, whereas the topic-aware semantic score serves as a supporting signal. For LongMemEval, the optimal weights are 0.5, 0.4, and 0.1, which reflects relatively stronger reliance on lexical features.

Both optimal configurations assign non-zero weight to the topic-aware semantic score. Even though its individual contribution is relatively modest, it supplies complementary information for candidate ranking. Top-performing weight combinations cluster around the optimal point, which implies certain robustness against minor weight perturbations. The discrepancy between the optimal weight settings across the two datasets further illustrates that a

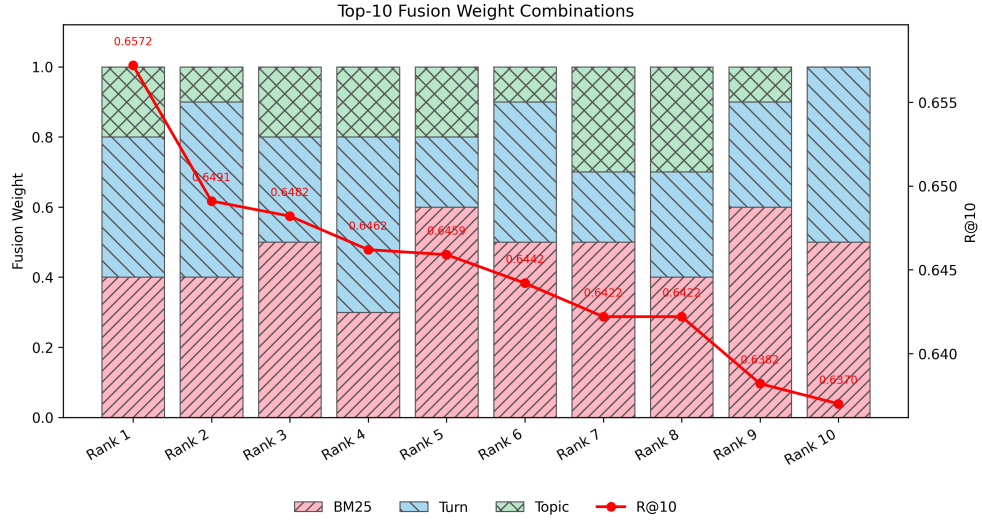


Figure 3: Top-10 ranking-weight combinations on LoCoMo

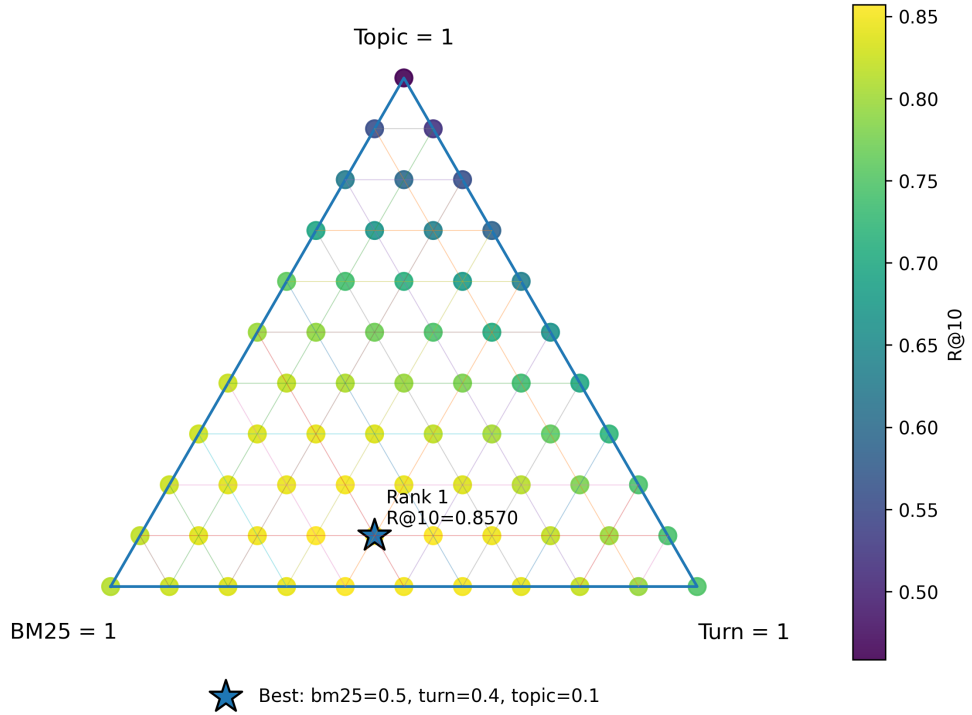


Figure 4: Grid-search results for the three ranking weights on LongMemEval colored by R@10

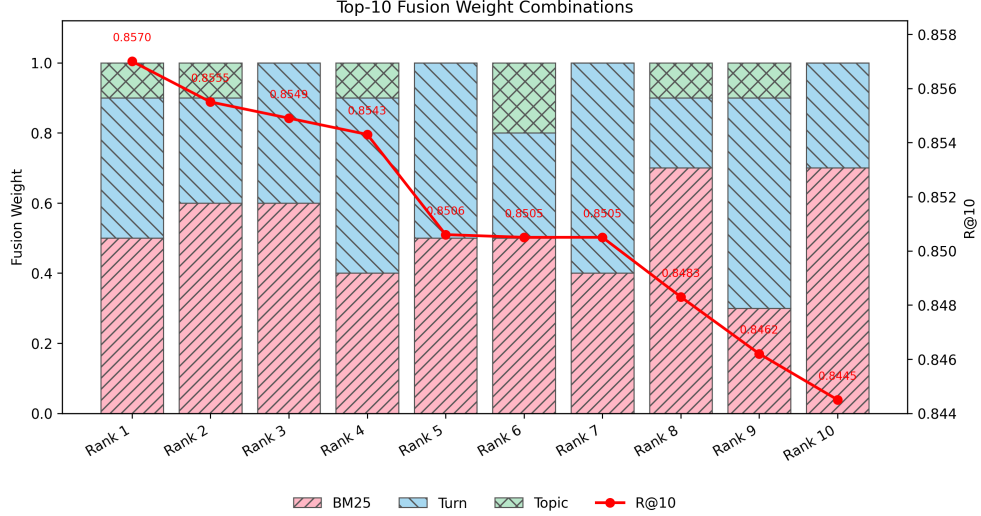


Figure 5: Top-10 ranking-weight combinations on LongMemEval

fixed weighting scheme cannot fully adapt to different data distributions and question categories.

4.6. Runtime Efficiency Analysis

Table 3: Experimental comparison of preprocessing time and average per-query processing time on two datasets

Method	LoCoMo		longmemeval_s_cleaned	
	Offline construction time (h)	Average online query latency (s)	Offline construction time (h)	Average online query latency (s)
MemGAS [17]	0.204	0.762	32.633	1.80
A-Mem [16]	5.389	0.9	209.734	6.22
SeCom [15]	0.329	3.43	1.179	1.95
HG-Mem	0.446(turn)	0.32	68.6	0.43

Table 3 compares the offline construction time and average online query latency of different methods on LoCoMo and LongMemEval. In particular, we use 32 GB memory of the MetaX C500 GPU for the preprocessing of A-Mem due to VRAM constraints. Since offline construction incurs substantially higher computational overhead, it is reported in hours. The average online query latency accounts only for the retrieval stage and is measured in seconds. For LoCoMo, we directly adopt the annotated session summaries provided by the dataset. For LongMemEval, HG-Mem is required to generate session summaries, which explains the discrepancy in preprocessing time

across the two datasets. Nevertheless, once the hierarchical memory structure is built, online inference only involves lexical retrieval, vector-similarity computation, candidate expansion, and linear ranking, with no dynamic memory updates or online LLM reasoning steps. Therefore, HG-Mem fits well for scenarios where the long-term memory store receives repeated queries, such that its one-time offline overhead can be amortized across successive query requests.

4.7. Ablation Study

To investigate the contribution of each candidate-recall path, we build three single-path ablation variants. Abl_BM25 retrieves candidate turns solely via BM25 and ranks candidates using the BM25 lexical score. Abl_SemSession relies exclusively on the session path: it selects relevant sessions based on semantic similarity between the query and session representations, then expands all turns contained within these sessions. Abl_SemTopic uses only the topic-cluster path: it picks relevant clusters according to semantic similarity between the query and topic-cluster representations and expands their corresponding turns.

Table 4: Overall metric performance of ablation variants and our method on two datasets (all metrics are reported as percentages).

Method	LoCoMo						longmemeval_s_cleaned					
	R@1	R@3	R@5	R@10	MRR	F1-score	R@1	R@3	R@5	R@10	MRR	F1-score
Abl_BM25	29.93	44.95	51.89	58.72	42.38	32.71	35.26	60.69	69.83	81.20	65.93	42.22
Abl_SemSession	12.14	23.58	29.40	36.58	22.39	23.93	24.68	44.31	55.07	69.80	52.49	38.23
Abl_SemTopic	10.64	18.15	20.91	24.53	17.70	18.61	22.68	37.70	45.21	54.26	45.54	33.14
Our Method	30.06	49.44	56.88	65.72	45.41	35.78	35.32	61.80	72.83	85.70	66.80	44.25

Table 4 shows that BM25 is the most consistently effective among the three single-path variants. This confirms that entities and lexical cues within raw turns remain critical for locating evidence from long-term memory. The session and topic cluster paths yield weaker performance when operated in isolation. This suggests that high-level representations obtained via summarization or topic aggregation cannot fully retain specific names, times, numbers, and fine-grained event details.

The full HG-Mem model outperforms all single-path variants across all metrics. This demonstrates that different recall paths supply complementary evidence signals, and multi-signal ranking brings improvements to candidate ordering.

4.8. Analysis by Category

Table 5: Per-category Recall@5 (%) of ablation variants and our method on the LoCoMo dataset.

Category	Abl_BM25	Abl_SemSession	Abl_SemTopic	Our Method
single-hop	17.78	16.08	13.11	25.99
temporal	60.44	36.20	26.09	65.60
inference	23.26	18.07	13.64	29.40
multi-hop	59.55	35.33	24.71	65.50
adversarial	58.52	23.99	16.37	59.30

Using the category field provided by LoCoMo, we partition questions into five categories: single-hop, temporal, inference, multi-hop, and adversarial. We compute Recall@5 for each question category. As shown in Table 5, HG-Mem achieves the best performance on single-hop, temporal, inference, and multi-hop questions. It exhibits particularly stable performance for questions that demand contextual associations or multiple pieces of historical evidence. This indicates that HG-Mem helps mitigate evidence omissions originating from direct turn-level matching. For adversarial questions, HG-Mem yields comparable results to the best single-path variant. This observation reveals no obvious performance drop under distractor interference, yet it does not prove dedicated robustness against adversarial perturbations.

Table 6: Per-category Recall@5 (%) performance of ablation variants and our method on the longmemeval_s_cleaned dataset.

Category	Abl_BM25	Abl_SemSession	Abl_SemTopic	Our Method
single-session-user	90.62	73.44	62.50	94.53
single-session-assist	85.71	83.93	76.79	89.29
knowledge-update	81.02	56.94	41.20	83.10
temporal-reasoning	68.54	43.21	36.16	69.95
multi-session	52.67	43.68	32.65	55.87
single-session-preference	46.11	57.22	51.11	54.44

As shown in Table 6, we further analyze Recall@5 results of HG-Mem across six scenarios in LongMemEval: single-session-user, single-session-assist, single-session-preference, knowledge-update, temporal-reasoning, and multi-session.

HG-Mem achieves the best performance in five of the six scenarios, although Abl_SemSession performs better in the single-session-preference scenario. This suggests that preference-related questions rely more heavily on session-level summaries. It further demonstrates that a single fixed weighting scheme cannot yield optimal performance for every question type.

4.9. Discussion and Sources of Error

The above experimental results reveal several key sources of retrieval error in HG-Mem. First, session summaries compress complete dialogue history. If key entities, temporal information, or user preferences are omitted during compression, the session path may fail to rank the correct session with high priority. This issue is particularly prominent for questions that rely on fine-grained factual details. Second, the quality of LLM-generated turn topics directly governs topic-cluster construction. Overly general topic descriptions may merge semantically distinct turns into a single cluster. In contrast, overly specific descriptions can split turns describing the same event into separate clusters, which degrades clustering consistency and retrieval accuracy. Finally, the current HG-Mem framework lacks explicit mechanisms for memory forgetting, conflict resolution, and state updating. When users hold temporally conflicting preferences or update personal states across different sessions, both outdated and up-to-date evidence may be retrieved. The model cannot automatically identify which memory entries remain valid for the current query, thereby introducing retrieval ambiguity and downstream answer errors.

5. Conclusions and Future Work

This paper addresses two interrelated limitations within long-term conversational memory: the lack of explicit structural connections across different memory granularities, and the challenge of leveraging high-level semantics to locate original turn-level evidence. We propose HG-Mem, a hierarchical graph and multi-path retrieval based framework tailored for long-term memory management. It organizes dialogue history into a hierarchical memory graph consisting of session, topic-cluster, and turn nodes. Three-path candidate recall together with multi-signal ranking returns original turns as evidence. Experiments conducted on LoCoMo and LongMemEval demonstrate that HG-Mem enhances relevant-evidence coverage and overall rank-

ing quality for long dialogue history, while maintaining low online retrieval overhead.

Several directions are worthy of future exploration. First, the existing hierarchical memory graph comprises session, topic-cluster, and turn nodes. Time and event nodes can be incorporated to better model temporal evolution and event-level dependencies. Second, HG-Mem currently performs candidate ranking with fixed weights for the BM25 lexical score, turn-level semantic score, and topic-aware semantic score. Dynamically adjusting these weights according to question types may bring performance improvements for preference-oriented, temporal, and cross-session questions. Third, the framework still lacks built-in mechanisms for memory forgetting, conflict detection, and factual state updating. When dialogue history contain state transitions or contradictory facts, both outdated and updated evidence may be retrieved, and the system cannot judge the valid time scope for each memory entry. Finally, HG-Mem could be extended toward multimodal long-term memory, tool-augmented agents, and open-domain personalized interactions to further evaluate its generalizability and practical applicability.

References

- [1] Z. Tan, J. Yan, I.-H. Hsu, R. Han, Z. Wang, L. Le, Y. Song, Y. Chen, H. Palangi, G. Lee, A. R. Iyer, T. Chen, H. Liu, C.-Y. Lee, T. Pfister, In prospect and retrospect: Reflective memory management for long-term personalized dialogue agents, in: Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Association for Computational Linguistics, Vienna, Austria, 2025, pp. 8416–8439. doi:10.18653/v1/2025.acl-long.413.
URL <https://aclanthology.org/2025.acl-long.413/>
- [2] K. T.-i. Ong, N. Kim, M. Gwak, H. Chae, T. Kwon, Y. Jo, S.-w. Hwang, D. Lee, J. Yeo, Towards lifelong dialogue agents via timeline-based memory management, in: Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers), Association for Computational Linguistics, Albuquerque, New Mexico, 2025, pp. 8631–8661. doi:10.18653/v1/2025.naacl-long.435.
URL <https://aclanthology.org/2025.naacl-long.435/>
- [3] J. Kang, M. Ji, Z. Zhao, T. Bai, Memory OS of AI agent, in: Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, Association for Computational Linguistics, Suzhou, China, 2025, pp. 25961–25970. doi:10.18653/v1/2025.emnlp-main.1318.
URL <https://aclanthology.org/2025.emnlp-main.1318/>
- [4] Z. Dai, Z. Yang, Y. Yang, J. Carbonell, Q. Le, R. Salakhutdinov, Transformer-XL: Attentive language models beyond a fixed-length context, in: Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics, Association for Computational Linguistics, Florence, Italy, 2019, pp. 2978–2988. doi:10.18653/v1/P19-1285.
URL <https://aclanthology.org/P19-1285/>
- [5] N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, P. Liang, Lost in the middle: How language models use long contexts, Transactions of the Association for Computational Linguistics 12 (2024) 157–173.
URL <https://aclanthology.org/2024.tacl-1.9/>

- [6] O. Goldman, A. Jacovi, A. Slobodkin, A. Maimon, I. Dagan, R. Tsarfaty, Is it really long context if all you need is retrieval? towards genuinely difficult long context NLP, in: Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, Association for Computational Linguistics, Miami, Florida, USA, 2024, pp. 16576–16586. doi:10.18653/v1/2024.emnlp-main.924.
URL <https://aclanthology.org/2024.emnlp-main.924/>
- [7] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, D. Kiela, Retrieval-augmented generation for knowledge-intensive nlp tasks, in: Advances in Neural Information Processing Systems, Vol. 33, Curran Associates, Inc., 2020, pp. 9459–9474.
URL https://proceedings.neurips.cc/paper_files/paper/2020/file/6b493230205f780e1bc26945df7481e5-Paper.pdf
- [8] K. Guu, K. Lee, Z. Tung, P. Pasupat, M. Chang, Retrieval augmented language model pre-training, in: Proceedings of the 37th International Conference on Machine Learning, Vol. 119 of Proceedings of Machine Learning Research, PMLR, 2020, pp. 3929–3938.
URL <https://proceedings.mlr.press/v119/guu20a.html>
- [9] P. Xu, W. Ping, X. Wu, L. McAfee, C. Zhu, Z. Liu, S. Subramanian, E. Bakhturina, M. Shoeybi, B. Catanzaro, Retrieval meets long context large language models, in: International Conference on Learning Representations, Vol. 2024, 2024, pp. 49569–49584.
URL https://proceedings.iclr.cc/paper_files/paper/2024/file/d75f29006df67df084e6586f1cb8458c-Paper-Conference.pdf
- [10] B. Jin, J. Yoon, J. Han, S. Arik, Long-context llms meet rag: Overcoming challenges for long inputs in rag, in: International Conference on Learning Representations, Vol. 2025, 2025, pp. 37784–37822.
URL https://proceedings.iclr.cc/paper_files/paper/2025/file/5df5b1f121c915d8bdd00db6aac20827-Paper-Conference.pdf
- [11] W. Yang, J. Xiao, H. Zhang, Q. Zhang, Y. Wang, B. Xu, Coarse-to-fine grounded memory for LLM agent planning, in: Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, Association for Computational Linguistics, Suzhou, China, 2025, pp.

13029–13056. doi:10.18653/v1/2025.emnlp-main.659.
URL <https://aclanthology.org/2025.emnlp-main.659/>

- [12] Y. Zhang, W. Yuan, Z. Jiang, Bridging intuitive associations and deliberate recall: Empowering LLM personal assistant with graph-structured long-term memory, in: Findings of the Association for Computational Linguistics: ACL 2025, Association for Computational Linguistics, Vienna, Austria, 2025, pp. 17533–17547. doi:10.18653/v1/2025.findings-acl.901.
URL <https://aclanthology.org/2025.findings-acl.901/>
- [13] A. Maharana, D.-H. Lee, S. Tulyakov, M. Bansal, F. Barbieri, Y. Fang, Evaluating very long-term conversational memory of LLM agents, in: Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Association for Computational Linguistics, Bangkok, Thailand, 2024, pp. 13851–13870. doi:10.18653/v1/2024.acl-long.747.
URL <https://aclanthology.org/2024.acl-long.747/>
- [14] D. Wu, H. Wang, W. Yu, Y. Zhang, K.-W. Chang, D. Yu, Longmemeval: Benchmarking chat assistants on long-term interactive memory, in: International Conference on Learning Representations, Vol. 2025, 2025, pp. 86809–86836.
URL https://proceedings.iclr.cc/paper_files/paper/2025/file/d813d324dbf0598bbdc9c8e79740ed01-Paper-Conference.pdf
- [15] Z. Pan, Q. Wu, H. Jiang, X. Luo, H. Cheng, D. Li, Y. Yang, C.-Y. Lin, H. V. Zhao, L. Qiu, J. Gao, Secom: On memory construction and retrieval for personalized conversational agents, in: International Conference on Learning Representations, Vol. 2025, 2025, pp. 91851–91885.
URL https://proceedings.iclr.cc/paper_files/paper/2025/file/e56f394bbd4f0ec81393d767caa5a31b-Paper-Conference.pdf
- [16] W. Xu, Z. Liang, K. Mei, H. Gao, J. Tan, Y. Zhang, A-mem: Agentic memory for llm agents, in: Advances in Neural Information Processing Systems, Vol. 38, Main Conference, Curran Associates, Inc., 2025, pp. 17577–17604. doi:10.52202/085713-0593.
URL https://proceedings.neurips.cc/paper_files/paper/2025/file/19909c36f51abc4856b4560aff3d36d6-Paper-Conference.pdf

- [17] D. Xu, Y. Wen, P. Jia, Y. Zhang, W. Zhang, Y. Wang, H. Guo, R. Tang, X. Zhao, E. Chen, T. Xu, From single to multi-granularity: Toward long-term memory association and selection of conversational agents, in: International Conference on Learning Representations, Vol. 2026, 2026, pp. 91581–91617.
URL https://proceedings.iclr.cc/paper_files/paper/2026/file/93e444b4a1a64815631c3c74977ee200-Paper-Conference.pdf
- [18] W. Zhong, L. Guo, Q. Gao, H. Ye, Y. Wang, Memorybank: Enhancing large language models with long-term memory, in: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 38, Vancouver, Canada, 2024, pp. 19724–19731. doi:10.1609/aaai.v38i17.29946.
URL <https://ojs.aaai.org/index.php/AAAI/article/view/29946>
- [19] C. Packer, S. Wooders, K. Lin, V. Fang, S. G. Patil, I. Stoica, J. E. Gonzalez, Memgpt: Towards llms as operating systems (2024). arXiv:2310.08560.
URL <https://arxiv.org/abs/2310.08560>
- [20] Q. Wang, Y. Fu, Y. Cao, S. Wang, Z. Tian, L. Ding, Recursively summarizing enables long-term dialogue memory in large language models, Neurocomputing 639 (2025) 130193. doi:<https://doi.org/10.1016/j.neucom.2025.130193>.
URL <https://www.sciencedirect.com/science/article/pii/S0925231225008653>
- [21] D. Edge, H. Trinh, N. Cheng, J. Bradley, A. Chao, A. Mody, S. Truitt, D. Metropolitansky, R. O. Ness, J. Larson, From local to global: A graph rag approach to query-focused summarization (2025). arXiv:2404.16130.
URL <https://arxiv.org/abs/2404.16130>
- [22] B. J. Gutiérrez, Y. Shu, Y. Gu, M. Yasunaga, Y. Su, Hipporag: Neurobiologically inspired long-term memory for large language models, in: Advances in Neural Information Processing Systems, Vol. 37, Curran Associates, Inc., 2024, pp. 59532–59569. doi:10.52202/079017-1902.
URL https://proceedings.neurips.cc/paper_files/paper/2024/file/6ddc001d07ca4f319af96a3024f6dbd1-Paper-Conference.pdf
- [23] J. H. W. Jr., Hierarchical grouping to optimize an objective function, Journal of the American Statistical Association 58 (301) (1963) 236–

244. doi:10.1080/01621459.1963.10500845.
 URL <https://www.tandfonline.com/doi/abs/10.1080/01621459.1963.10500845>
- [24] S. Robertson, S. Walker, S. Jones, M. M. Hancock-Beaulieu, M. Gatford, Okapi at trec-3, in: Overview of the Third Text REtrieval Conference (TREC-3), Gaithersburg, MD: NIST, 1995, pp. 109–126.
 URL <https://www.microsoft.com/en-us/research/publication/okapi-at-trec-3/>
- [25] N. Reimers, I. Gurevych, Sentence-BERT: Sentence embeddings using Siamese BERT-networks, in: Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP), Association for Computational Linguistics, Hong Kong, China, 2019, pp. 3982–3992. doi:10.18653/v1/D19-1410.
 URL <https://aclanthology.org/D19-1410/>
- [26] G. V. Cormack, C. L. A. Clarke, S. Buettcher, Reciprocal rank fusion outperforms condorcet and individual rank learning methods, in: SIGIR '09: Proceedings of the 32nd international ACM SIGIR conference on Research and development in information retrieval, New York, NY, USA, 2009, pp. 758–759.
 URL <http://doi.acm.org/10.1145/1571941.1572114>